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EDUCATION

Institut Teknologi Bandung, Indonesia
Master of Science in Computational Sciences
· GPA: 3.98/4.0

9 September 2024 – 12 August 2026

Institut Teknologi Bandung, Indonesia
Bachelor of Science in Astronomy
· GPA: 3.25/4.0

17 September 2020 – 24 August 2024

EXPERIENCE

Research Intern – ASPIRE Program
University of Amsterdam

June 2026 – August 2026
Amsterdam, The Netherlands

- Re-engineered a fixed-timestep physics simulation into an event-driven adaptive-situation pipeline cutting runtime from ~ 1 –2 days to ~ 5 minutes while preserving numerical accuracy.
- Designed a memory-bounded architecture with streaming compressed (HDF5) output, keeping RAM usage independent of simulation length, and derived the geometric and limb-darkening models from first principles (vectorized NumPy/Shapely) to generate synthetic signals for controlled hypothesis testing.

Graduate Researcher – Machine Learning Engineer
Institut Teknologi Bandung

February 2025 – August 2026
Bandung, Indonesia

ENTRAP (*semi-supervised outlier recovery*):

- Designed a model-agnostic layer that re-scores an unsupervised model's rejected outliers via topological consistency instead of a fixed density cutoff; reached 1.0 precision and $>95\%$ recall on 99 labeled synthetic datasets, and $+13\%$ to $+94\%$ recall gains on 10 real-world datasets (8 of 10 cases).
- Replaced one global hyperparameter with a per-cluster estimator derived from local intrinsic dimensionality, removing manual threshold tuning entirely.
- Optimized the pipeline in Cython with parallel processing and an opt-in memory-mapped backend, achieving sub-quadratic $O(N^{1.84})$ scaling and support for $>50\text{K}$ -point datasets under constrained RAM; packaged with a scikit-learn-compatible API [code].

GUMM (*unsupervised mixture modeling*):

- Built an EM-based mixture model with an explicit noise component and benchmarked it against scikit-learn's Gaussian Mixture (3 scenarios, 20 seeds, Wilcoxon test): significantly higher precision and ARI in all 3 ($p < 0.001$; ARI 0.65 vs -0.02 on the hardest case), trading ~ 135 – $270\times$ fit time for the accuracy gain.
- Validated on a 23K-point real-world dataset by cross-checking against an independent diagnostic unused in training, confirming correct cluster recovery where the baseline model produced a clear artifact from its missing noise component [code].

Laboratory Assistant – Machine Learning & Data Mining
Institut Teknologi Bandung

February 2025 – June 2026
Bandung, Indonesia

- Taught and mentored ~ 20 – 30 graduate students across two courses (Advanced Machine Learning, Data Mining), covering the full workflow from data preprocessing to modeling with neural networks, CNNs, SVM, KNN, and K-Means.
- Audited student projects end-to-end as a QA reviewer, validating exploratory analysis, model architecture, and evaluation results to catch methodological errors.
- Designed a hands-on data analysis project (NumPy, Pandas) and standardized data validation workflows and documentation used across cohorts.

Data Analyst – Signal Processing & Statistical Modeling
Institut Teknologi Bandung

July 2023 – April 2026
Bandung, Indonesia

- Reconstructed spatial signals from raw frequency-domain data and built a sigma-clipping pipeline to denoise high-dimensional spectral data before analysis.
- Fit noisy 1D signals and a physics-constrained multi-wavelength model via non-linear least-squares (Levenberg–Marquardt, `lmfit`), propagating uncertainty to quantify confidence in every derived result.
- Decomposed 3D volumetric data cubes into statistical feature maps, surfacing a localized anomaly invisible in the aggregated signal.
- Published findings as first author in *Monthly Notices of the Royal Astronomical Society*, a top-tier peer-reviewed journal [paper].

PROJECTS

ENTRAP: Entropy-Based Topological Rescue of Ambiguous Points

2026

Master's Thesis, Institut Teknologi Bandung

Python, Cython, C, Numba

- Designed a hybrid topology-based algorithm that rescues ambiguous points labeled as noise by HDBSCAN, targeting more complete open cluster membership identification in Gaia DR3 data.
- Ranked HDBSCAN noise points by a Mahalanobis-based Gaussian likelihood distance to each cluster and accepted candidates up to the knee of the sequential H_0 persistence-entropy curve (Ripser, Knead), with TwoNN intrinsic-dimension estimation adapting the local neighborhood size; exposed a scikit-learn-style ENTRAP estimator supporting more than 10 distance metrics and per-cluster diagnostic plots.
- Implemented three functionally equivalent builds (Numba JIT, Cython, and C with an automatic NumPy fallback) to support environments with and without a compiler toolchain [Python] [Cython] [C] [Thesis].

GUMM: Gaussian-Uniform Mixture Model

2026

Open-Source Python Package

Cython, OpenMP

- Built an unsupervised membership classifier that separates K Gaussian cluster components from a uniform background via Expectation-Maximization, with a fully automatic membership threshold (rotation-elbow method, optionally refined by a Ripley's K Monte Carlo test); no manual cutoff tuning required [GitHub] [Paper].
- Parallelized the two compute-intensive regions, the Monte Carlo Ripley's K simulations and the weighted-covariance M-step, using OpenMP; released under the MIT license with a scikit-learn-style API (`fit`, `predict`, `predict_proba`, `score_samples`).

Kepler-221 Transit Timing Variation Pipeline

2026

ASPIRE Program, University of Amsterdam

Python, AMUSE, Huayno

- Built, within a three-person research team, an end-to-end pipeline that converts AMUSE N-body output into synthetic transit light curves, closing a gap for which no existing package was available; used to test whether an undiscovered outer planet explains the broken resonance of Kepler-221 d [GitHub].
- Combined adaptive-timestep Huayno integration, sky-plane geometric projection, and Mandel-Agol limb-darkening models; preliminary results show non-zero TTV residuals for planet e, consistent with an unseen perturber.

EASLT: Enhanced Automatic Shifted Log Transformer

2025

Open-Source Python Package (PyPI)

Python, Numba, SciPy

- Published a scikit-learn-compatible transformer on PyPI that automatically normalizes skewed data by selecting shifted-log parameters through Monte Carlo weight optimization over Anderson-Darling, skewness, kurtosis, and stability scores, handling zero and negative values without manual tuning [GitHub].
- Accelerated the parameter search with Numba and added adaptive outlier winsorization, quality-control thresholds, and full inverse-transform support for use inside standard ML pipelines.

Brain Tumor Detection with Topological Data Analysis

2025

Independent Research Project

PyTorch, torch-topological

- Designed a three-branch MRI classifier (CNN, topology, and fusion), trained in two stages, that fuses VGG19 features with topological descriptors: Grad-CAM++ attention maps are sampled into point clouds, converted to persistence images through Vietoris-Rips persistent homology (H_0 , H_1), and concatenated with CNN features for binary tumor classification [GitHub].
- Exposed attention-guided topological signatures to support model interpretability and stabilized training by isolating the topology branch from the CNN through gradient detachment.

Speech Denoising with a Recurrent-Convolutional Neural Network

2025

Independent Project

PyTorch, librosa, torchaudio

- Developed a spectrogram-mask speech denoising model based on a Recurrent-Convolutional Neural Network (R-CNN), trained on three public noise and speech-activity datasets; reached a PESQ of 2.445 and a STOI of 0.922 on a demonstration sample, indicating acceptable perceptual quality and excellent speech intelligibility [GitHub].

TECHNICAL SKILLS

Languages: Python, C/C++, Cython, SQL, L^AT_EX, HTML, CSS, JavaScript

Astronomical Software & Libraries: AMUSE, Galpy, Astropy, ASteCA, PyVO (TAP/ADQL/SQL), CASA, Spectral-Cube

Machine Learning Software & Libraries: PyTorch (torchvision, torchaudio), TensorFlow, Keras, Scikit-Learn, Ripser and torch-topological (TDA), HDBSCAN, pyABC (Bayesian inference), Knead, librosa, XGBoost

General Software & Libraries: NumPy, SciPy, Pandas, Matplotlib, Seaborn, Numba, Joblib, Shapely, h5py, OpenMP, MPI (mpi4py, mpich), lmfit, Scikit-Learn

Developer Tools: Git, Linux/Windows, VS Code, Jupyter, Google Colab, PyPI Packaging, Scikit-Learn API Design

CERTIFICATION

freeCodeCamp

August 2024

Machine Learning with Python – Developer Certification

- Completed approximately 300 hours of coursework covering TensorFlow and Keras for neural networks, gradient descent techniques, classification with neural networks, and natural language processing with Scikit-Learn and TensorFlow.

DQLab

July 2021 – August 2021

Data Science Training Program

- Python for Data Professional
- Exploratory Data Analysis with Python for Beginner.
- Data Visualization with Python Matplotlib for Beginner
- Eksplorasi dan Analisis Data COVID-19 Indonesia using Python.

ACHIEVEMENTS

Dean's Award for Academic Achievement

February 2024

Faculty of Mathematics and Natural Sciences, Institut Teknologi Bandung

- Awarded for Semester I 2023/2024 to undergraduate students achieving a semester GPA of at least 3.50 with a minimum grade of B in every course.